Annex F1:
IT & VAS Managed Services

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1. Introduction

The scope of Managed Services is for the Contractor to engage with the Purchaser in an outsourcing project to define and implement the Future Mode of Operation for ABCD OpCo IT & VAS. This project will involve a significant demerger of some of our operations and transfer to one successful technology partner.

Our aspirations include:

1. Enhancing "Time to market" capability and responsiveness to launch new services build and deliver IT & VAS Products and Services and do so in an agile manner;
2. Lead, or equal-lead, the market on IT & VAS QoS to complement segmented customer experience; and
3. Deliver best-in-class cost structure for all our IT & VAS operations and infrastructure.

The aspirational operating model we are seeking is based on the principle of a "new entrant" scenario; in other words if we were entering the OpCoi market for the first time and wanted to serve 2 million pre & postpaid clients, we would have to set up a certain capability for our IT & VAS operations. This is the "new entrant" operating FMO model we are seeking. Of course ABCD OpCo has a legacy and need to migrate; however, within a certain period, within 3 months of contract conclusion, we are keen to operate in such a way.

ABCD OpCo is committed to implement such an operating model, and our decision will be based on:

A. How well your responses satisfy the 3 key aspirations above.
B. Contractual Terms and Conditions.
C. Best transition strategy which takes into accounts our people and minimum if any disruption to Quality of Service.

2. Compliance Matrix

2.1. Requirements Definition and Compliance

- The Compliance matrix defines the functional/non-functional requirements for equipment/ software to be offered and services to be rendered.
- The contractor shall respond to the requirement specifications clause by clause, using a compliance summary statement, and will be submitted as described in "Annex F5-IT- VAS-Managed-Services-Compliance-Matrix v1" Excel file.

2.2. Compliance Definition

The compliance column of the compliance summary statement shall contain one and only one of the following acceptable compliance statements.

Compliance Statement	Meaning
Fully compliant	The offered equipment/solution/service is fully compliant with the requirement of the clause
Not compliant	The offered equipment/solution/service does not strictly comply with the requirements of the clause.

- For purpose of compliance verification, a document and page number cross-reference shall be given (where applicable) in the reference column of the compliance summary statement.
- Any clause marked with "fully compliant" in the contractor compliance summary statement is binding on the successful tender to supply equipment/solution/service that meets the requirements of that clause. This applies for the lifetime of any awarded contract.

3. Requirements for IT & VAS Managed Services

The below sections details the ABCD OpCo requirements for IT & VAS Managed Services.

The contractor shall cover the following areas within IT & VAS managed services scope.

- ✓ NOC IT & VAS System Monitoring (Level 1)
- ✓ End user Help Desk Support.
- ✓ Applications Management Service (**AMS**) such as:
  - BSS modules Operation & Management
  - CRM Operation & Management
  - Provisioning Operation & Management
  - Mediation Operation & Management
  - Data warehousing/ Business Intelligence Operation & Management
  - Enterprise Service Bus (ESB)Middleware operation & Management
  - Internet Portal Operation & Management
  - ERP Operation & Management
  - VAS platforms
- ✓ Infrastructure Management Service **(IMS)** such as **:**
  - Servers Operation & Management
  - Storage Operation & Management
  - Databases Operation & Management
  - Virtualization operations, management & provisioning
  - Backup & Archiving Operations & management
- ✓ IT & VAS Service Assurance such as:
  - BMC Remedy
  - HP OMI
  - HP uCMDB
  - HP NNMi
  - HP BSM
  - HP BPM
  - HP RUM
  - HP Diagnostic
  - HP UD
  - HP operations analytics

3.1. NOC IT & VAS System Monitoring (Level 1)

ABCD OpCo is looking to implement the best of bird SOC (Service Operation Center) and aligned with industry standards (ITIL V3 & eTOM). Contractor shall perform the following activities but not limited to:

- Monitor IT & VAS infrastructure, platforms and applications.
- Monitor all interconnections between IT, Transmission (IP MPLS), Core Network, VAS Systems, Internet, and all other connectivity.
- Monitor back up completion of File systems, and database backups through the automated backup software.
- Incidents Management Level 1 supports.
- Assist in generating daily/weekly/monthly availability reports.
- Run daily batch jobs if needed.

3.2. End user Help Desk Support

The “End User Help Desk Support Service” shall support the business users in ABCD OpCo and the other

remote sites (Sales offices or Branches) with the following service specific activities but not limited to:

- Software installation
  1. Office installation: all PC's will be installed office suite like (word, excel, outlook). This may include adjust setting like configuring email account.
  2. Operating installation: Setup windows on PC's, with all required setting and polices along with Antivirus.
  3. Other Program installation: install any program or application that may request from end user and comply with IT policy.

- Software troubleshooting
  1. Office troubleshooting: may include configure, repair and reinstall any of office suites.
  2. Login error (disk full): recover; delete any unnecessary files that may affect PC performance.
  3. Login problem: may include repair OS, change some setting.
  4. Browsing problem: may include repairing IE and fixing any related issues (add-ons, plug-ins).
  5. Antivirus issues: may include update virus definition or repairing current installation and remove any viruses, worms, Trojans.

- Hardware installation:
  1. PC setup: may include setup new PC's or moving existing PC's to another location.
  2. Device Check: checking existing Hardware for the purpose of replacement.
  3. Printer installation: may include installation and configuration of printers also installing drivers on the selected PC's.
  4. Scanner installation: may include scanner installation along with required driver on the PC's connected to it.
  5. Printer ink: changing printer ink

- Other Activities:
  1. Provide troubleshooting of network connectivity and server issues.
  2. Proactively assisting with help desk duties, supporting printers, hardware, software and related technical issues.
  3. Installation and Configuration of Operating Systems on Desktops as well as Servers and all other ABCD OpCo related day to day software.
  4. Configuration Setup and Troubleshooting of Email client.
  5. Configuration and setup of Hardware such Servers, PC, Network Printers, Connect Cards (3G, GPRS, Wireless devices and network connectivity through mobile devices.
  6. Disaster Recovery and Backing up of user data on to removable medium.

7. Detecting, diagnosing and correcting computer hardware, software, and peripheral errors.
8. Keep track on the entire H/W and S/W ABCD OpCo inventory
9. Replace any hardware device which is not working
10. Ability to perform partition, drive space allocation and understand NTFS and FAT partitioning.
11. Onsite support provided by contractor is only limited to ABCD OpCo office, and retail shops.
12. DNS & file server should be added
13. Printers and peripherals support should be added
14. Printers & peripherals support should be added

- Messaging Operation & Management
  1. Manage and maintain email database and e-mail queue.
  2. Administer Email user (Creation, Deletion, and Access Privilege.)
  3. Define threshold for the inbound and outbound email queue.
  4. Provide threshold limits for performance counter, mail queue and backup for messaging infrastructure.
  5. Support customer mailbox connectivity as per the requirements.
  6. Maintain mail flow and database.
  7. Perform proactive fault detection and diagnostic procedures.
  8. Participate in the review and understanding of email server performance and implication of providing additional hardware.
  9. Forecast capacity and database storage requirements and provide appropriate storage to meet capacity projections
  10. Provide application integration and administration, policy management, data backup, restore and purging, application patching and update, user provisioning and management, spam mail control and Level 2 technical support for Microsoft Exchange.

- Microsoft Technologies Operation & Management:

  1. Administer all in-scope MS Windows servers for the end users
  2. Support of Window Server and all in-scope Microsoft products and Blackberry BES running in ABCD OpCo.
  3. Tuning for the Windows Enterprise Systems.
  4. Maintaining security of Windows servers.
  5. First and Second line support for Exchange, Active Directory, DNS, DHCP servers…
  6. Monitor, control and manage recovery from any unplanned outage.
  7. Apply patches to servers as and when required and as per the change management procedure.
  8. Install and update anti-virus software
  9. Maintain licensing issues related to the server applications
  10. Provide technical recommendations with relation to the Microsoft products.
  11. Optimize the operation environment.
  12. Perform daily health check (performance, CPU Usage, disk space, Memory usage) to ensure that the windows servers and applications are running efficiently.
  13. Authorize and provide recommendations on changes to the Windows-based servers.

14. Update or coordinate the update of all appropriate databases / documentation (problem tickets, Change Management documentation/reports, asset management system.).
15. Capacity planning of all Windows-based servers and its associated Microsoft applications.
16. Perform user administration on Windows-based servers, active directory, and Email system and restrict system access to authorized personnel's.
17. Supervise new installation and configuration on the current Windows Systems hardware and software, as per the change and release management procedures.
18. Coordinate with the vendors for support if needed according to signed maintenance contract.
19. Provide technical recommendations with relation to the Microsoft based products in order to enhance and optimize the operation environment.

3.3. Application Management Services – AMS

The Application Management Services shall include but not limited, Incident handling, Problem handling, patch management, Configuration management and Change Management. Contractor shall also make sure proper backups and archive was taken, applications performance tuning, purging of temporary tables/data and operations with the following models:

1. Performing regular health check to the systems and processes. Provide Level 2 support to all in-scope applications services.
2. Attend/respond/handle tasks assigned via service desk within the response and resolution agreed in the SLA.
3. Work with ABCD OpCo billing and applications department for functional area requiring support for systems setup and maintenance.
4. Assist ABCD OpCo in gathering the business user's requirements for any new functionality.
5. For external CRs initiated, participate in revising GAP document and provide recommendation on GAP suitability to requirement and any adverse impact on existing system functionality.
6. Testing of new patches / Software in on-site test environment and share the reports with ABCD OpCo for further processing.
7. Participate in UAT for any patch along with the ABCD OpCo Telecoms Business Users representing technical side of IT.
8. Deploy software for CRs and new releases: coordinating with implementation team in reviewing implementation plan and impact analysis and provide recommendation for deployment to ABCD OpCo IT Management. Coordinating with Infrastructure and Operations dept for any software / module deployment.
9. Help Business Users to understand the transactional flow and end results.
10. Report bugs in the application to 3rd line support perform the following regular tasks as per agreed reporting issues to ABCD OpCo IT Management through the IT Outsourcing Manager.
11. Performing Ad hoc Tasks: data clean up, reconciliation between systems and reporting related to ABCD OpCo operation/business enhancement.
12. Update ABCD OpCo management on potential risks and recommend risk mitigation measures after
13. Coordinate with 3rd line support if needed according to signed maintenance contract.

14. Proactively monitor different modules and report on their performance status.
15. Adhere to the Quality procedures that are agreed to be part of the service management scope
16. Check system running logs and give the analysis which support the conclusion of application is normal or abnormal.
17. Incident and problem handling
18. Receive, log, assess, prioritize, and resolve all support requests from system users, following the established support processes, related to applications as per design specifications.
19. Resolve processes and functional queries and incidents by assisting business users with system transactions, providing guidance and clarification on system behavior and performance as per design specifications. The queries may relate to applications: data, configuration, application operations or usage issues.
20. Perform investigation and root cause analysis on problems affecting system operation, and recommend resolution actions through the applicable application governance process.
21. Implement system configuration and code fixes required for problem resolution, including updates to the impacted documentation. Conduct integration testing, assist with user acceptance test execution and any associated data conversion activities.
22. Process system change requests as raised from authorized business users according to the applicable governance process, covering: change request logging and tracking, requirements analysis, solution design recommendation, and implementation subject to governance approval.
23. Conduct Governance Forums on periodic basis for review and approval of changes affecting applications.
24. Manage the implementation of approved change requests, including liaising with the relevant 3rd Party service providers or offshore development team.
25. Maintain design documentations up to date incorporating all changes from ongoing operation support.
26. Proactively monitor the overall technical environments and action based on monitoring alerts. Provide report on application performance status, targeted on weekly basis.
27. Perform periodic service performance analysis for identification of improvements.
28. Perform activities necessary to repair system to restore the functionality, availability and performance as required.
29. Support the application and deployment of software patches to the productive system landscape.
30. Perform release management for system changes specific to the applications.
31. Perform start-up / shut down of applications where required.
32. Support application level testing in conjunction with Infrastructure Team following Infrastructure Changes, e.g. Security Patch / DB Upgrades, where necessary.
33. Configuration changes or data correction to resolve application issues or to implement work around
34. Any changes to be deployed into Production will be approved through a standardized Change Management process to be defined and managed by Contractor Service Management function.
35. Reactive Services :
    - Incident and problem management for ABCD OpCo implementation
    - Work with ABCD OpCo , and third-party vendors to address implementation issues

36. Advisory Services :
    - Technical account management
    - Periodic solution assessments, adoption monitoring
37. Operational & Enhancement Services :
    - Out-task maintenance and administration activities
    - A complete responsibility for ongoing solution operations
    - Implement solution enhancements

- Below examples but not limited activities :

Solution Managed Services	Reactive Services	• Response to incidents within a defined service level objective • Solution support and defect correction
	Advisory Services	• Proactive evaluation of existing solution and recommendations for patches • Improvement of long-term stability and value of the entire solution by focusing on technical health, user adoption, and planning for continuous solution improvement
	Operational Services	• System administration, user account management • Technical support of the applications, including periodic checks to monitor application availability and response to alerts • Facilitation of the movement of changes (including defect fixes and patches) from development to test to production • Proactive review of log files and system usage to reduce potential for outages
	Enhancement Services	• Design, development and delivery of functional enhancements to the existing solution through Change Management Process.

New Projects	Implementation of new and major solution functionality	• Implementation of upgrades (major and minor point releases) to ensure that applications stay current and eligible for vendor technical support through Change Management Process. • Assist in Design, development, and deployment of new integrations between applications or major enhancement releases through Change Management Process.

Note (1): Refer excel file "**Annex F3-IT- VAS- Applications"** for list of IT & VAS systems that are in scope of the AMS.

3.3.1. BSS Modules Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the BSS applications such as:

1. Call Processing: Call Rating Setup (Rate plans, Tariff, Holiday setup, Suspended call monitoring, resolution and reporting, seeing the operation of all steps of Calls processing is error free, Perform any recoveries required.
2. Billing: Support for Bill Review, Ensuring Bill Run smoothly, any recoveries for additional requirements (ex: script to waive particular charges)/ bugs (Ex: Wrong allowance deduction), Extract and Archiving, Modifications to Extraction script.
3. Roaming Processing: Tariff Setup, Suspended call monitoring, resolution and reporting, seeing the operation of all steps of Calls processing is smooth and error free, perform any recoveries required, Investigation and assistance in Fraud cases.
4. Interconnect : Setup for different interconnect operators including the operators for SMS inter working / destinations and subsequent changes required, Ensuring processing of all interconnect CDRs, Production of Reports, Reconciliation and Investigation into discrepancies.

3.3.2. CRM Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the CRM applications such as:

1. System Authority: Users creation and assigning routine authorities, changing routine authorities, Field level authority setup as per business requirement.
2. Order monitoring and reconciliation. Monitor orders that have exceptional status, unassigned or fail to generate work order.
3. Maintain and check accounts opening & cancellation and service status changing. Monitor whether there is exceptional fluctuation or trend and resolve that
4. Modification of channel information, configuration of new channels, and deletion of old channels
5. Configure and maintain the inventory information, such as SIM card and phone number.
6. Maintain the customer centricity data and identity within CRM and assure consistency across CRM and all CRM interfaced system(provision and CRM,CRM and Billing)
7. Maintain the CRM performance and live cycle of the product catalogue and product life cycle

3.3.3. Provisioning Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the

provisioning applications such as:

1. Checking the Status of the Provision Processes
2. Assure that issued Customer Provisioning Orders are correct and complete.
3. Check that all necessary information about the Customer Provisioning Order (for example, type of product, install address, special requirements) is available.
4. Maintain the provisioning performance, logs, errors and failback operation
5. Maintain the integration agents with the southbound network elements and highlight and resolve any issues

3.3.4. Mediation Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the mediation applications such as:

1. Test and check the function of collecting the CDR from the network element (NE)
2. Monitoring collection process and check whether there are any error files
3. Check online mediation alarms
4. Generate mediation report
5. Handle abnormal CDR (error CDR re-processing and investigation)
6. Check mediation systems communication/link with other nodes
7. Analyze and troubleshoot online mediation errors/problems (if any)
8. Check online mediation systems capacity
9. Data processing - check and verify CDRs data (decoder, encoder), fields/records
10. Check and re-processing of error CDRs
11. Maintain the integration adapters towards the source system and assure the quality of the processes
12. Monitor and report any abnormal data collection patterns (losses, increase.)

3.3.5. Data warehousing & Business Intelligence Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the Data Warehousing/BI applications such as:

1. Auto refresh process support, troubleshooting, enhancement and following up any changes in BSS that may affect the process and providing the solution that keeps the process running properly.
2. Analyzing and Development of the Ad hoc queries required by ABCD OpCo Business users. Ensuring delivering business users right requirements within the agreed KPI.
3. Follow up on new requirements with customer and 3rd line support
4. Support and maintain ETL (Extract, transfer, Load) process.
5. Daily Patch loading: This daily batch involves the execution of ETL-JOBS and customized Process Flows
6. Data quality monitoring, report and escalate quality root cause to system owners
7. Manage failed ETL loading, error handling, resolve reference dependencies and re-loading process
8. Teradata Database Support and Maintenance

3.3.6. Enterprise Service Bus (ESB)/Middleware Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for the in-scope Middleware applications such as:

1. Middleware Operation and Management support services cover the overall status of web servers and web applications: the web server, web application, web cluster, web profiles are work well and stable.
2. Proactively monitor the transaction status and EJB Beans Available, Requests Waiting, and other transaction between other modules, such CRM stake.
3. Provide support to co-operate with other system for the interface integration.

4. Monitoring and analyze the root cause for log error and other problems such as the interruption. Perform investigation and root cause analysis on problems affecting transaction efficiency and recommend resolution actions.
5. Support the application and deployment of software update patches to the middleware.
6. Perform start-up / shut down of applications & servers where required.
7. Perform the capacity monitoring and provide the solution accordingly, such as the license status and the transaction limitation and space limitation.

3.3.7. Internet Portal Operation & Management:

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the Internet Portal such as:

1. Provide ongoing maintenance, initial triage, problem resolution and break-fix for the SiteCore application.
2. The support team will be involved in performing:
    - Manage activities / service requests that require code changes for defect fixing. Contractor shall have a well-defined process to manage the entire lifecycle for such changes ranging from design, development, testing to deploying into production. Any defect not pertaining to the configuration, customization done by Accenture would be escalated to the respective product vendors for resolution.
    - Additional Internet Portal specific activities include:
        - Resolve web layout errors due to application configuration.
        - Resolve content authoring issues due to application configuration.
        - Monitoring and maintenance of Site Core server threshold values.

3.3.8. ERP Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the ERP applications such as:

1. ERP Operation and Management support services cover the overall purchased ERP modules including interfaces to surrounding systems that together provide the complete ERP business capability to ABCD OpCo operations across key business functions of: Finance, Supply Chain, and Human Resources.
2. Escalate and liaise with Oracle as Application Vendor to resolve any SAP standard application defects, through the use of SAP OSS message.
3. Create and schedule batch jobs on user request; monitor jobs execution and follow-up action with job owners.

3.3.9. VAS Platforms

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the VAS Services/Platforms such as:

1. Daily Heath-check of VAS platforms/services. Analyze the logs/Alarms and resolve the issues (Storage capacity exceeded, NIC down, process hang-up, Hardware fault…)
2. Monitor the systems performance and raise the concern to VAS planning, in case of overload or outstanding behavior.
3. Open fault report with the vendor and ensure to solve the issue based the agreed SLA.
4. Execute the work-orders sent by VAS Planning and acknowledge the execution result.
5. Trait the subscriber's complaints from customer care and fix the bugs. If the change affects the business logic, VAS Planning should be informed to approve the update and send work order to fix the issue.
6. Assist with VAS planning in the integration of new VAS platforms.
7. Perform the ATP of new platforms/services jointly with VAS Planning.

8. Inform VAS Planning by weekly fault report
9. Maintain backup of the VAS platforms/Services.
10. Provide live inputs from the production platforms to VAS planning, when it is requested.
11. The contractor should be pro-active and inform VAS planning for any concern to avoid disaster

3.4. Requirements for IT & VAS Infrastructure and Platform Management Services – IMS

Contractor Infrastructure Managed services shall be responsible for operation and administration of all infrastructure and platforms components (Server, Storage, and Database..etc) hosted in ABCD OpCo Datacenters. Infrastructure and Platform Management Services comprises with the following models and activities.

3.4.1. Server Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the server operation such as:

1. Assist in the implementation of approved policies and procedures for operating system design, implementation and maintenance.
2. Provide input in the planning of operating systems upgrades and deployment.
3. Provide support in the implementation and user acceptance testing of operating systems.
4. Assist in the development of a proactive maintenance cycle for operating systems and automate preventive/routine maintenance on systems.
5. Perform daily operating system administration, including hardware & software upgrades.
6. Maintain backup of the operating system network.
7. Ensure optimal performance of all applications running on operating system environment
8. Perform daily systems checks and correct any issues.
9. Work with the User Support team to provide second level support to end users and ensure timely resolution of system problems.
10. Liaise with relevant unit/ team/function in carrying out all relevant activities.

3.4.2. Storage Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the storage operation such as:

1. Assist in the implementation of approved policies and procedures for data storage management and systems.
2. Participate in the sizing, designing, building, and delivering data storage system platforms.
3. Provide input in the development of information management lifecycle to achieve optimal mix of online, near online and offline data storage and quick user accessibility to the Storage Area Network and stored data.
4. Assist in the development of a proactive maintenance cycle for data storage systems and automate preventive/routine maintenance on data storage systems with detailed scheduled maintenance reports.
5. Maintain and update data storage systems.
6. Perform data backup and ensure that backed-up data can be restored as required.
7. Monitor data storage system performance and recommend adjustments to system parameters to ensure optimum performance and response time.
8. Ensure optimization of system logs on the data storage systems and proactively advise on capacity issues.
9. Keep system documentation updated at all times.
10. Assist users in defining their storage needs.
11. Assist in gathering requirements for additional storage requirements and provide support in the design and documentation of functional specifications for upgrades and expansions.
12. Liaise with User Support team to ensure data storage incidents are resolved efficiently and in a timely manner in compliance with Service Level Agreements.
13. Escalate incidents to third party vendors as required.
14. Assist to manage vendors providing services to the end user support function and ensure that agreed

SLAs are achieved.
15. Liaise with relevant unit/ team/function in carrying out all relevant activities.
16. Maintain and manage Storage Area Network (SAN)
17. Day to day storage provisioning
18. SRDF and storage point in time clones management & configuration
19. Support & configuration of Brocade enterprise SAN switches

3.4.3. Database Operation & Management

Contractor shall provide ongoing day-to-day operation and maintenance support for all functions of the database operation such as:

1. Assist in the definition of database architecture and design specifications
2. Work with IT vendors to install, configure, maintain, and upgrade enterprise database systems and related applications.
3. Provide support in functionality and acceptance testing during the implementation of enterprise database systems.
4. Ensure optimal performance of the organization's databases.
5. Regularly monitor and reconcile the performance of the databases to pre-defined performance thresholds and take appropriate actions where required.
6. Implement established policies and procedures on database management.
7. Maintain database security and data integrity through the use of access codes and efficient password system.
8. Perform regular data backup, and validate integrity of data backup.
9. Perform database automation and script writing.
10. Investigate and resolve incidents/ requests relating to enterprise databases, and escalate irresolvable faults/ issues to relevant vendors/ suppliers where appropriate.
11. Provide support in operating systems trouble shooting.
12. Ensure availability of accurate documentation of standards, guidelines and procedures for effectively managing and maintaining the databases.
13. Liaise with relevant unit/ team/function in carrying out all relevant activities.
14. Databases binaries installation and database instances creation should be part of the scope. That covers oracle database binaries, oracle grid binaries, packs enablement, data guard configuration
15. Solix database software management, maintenance & configuration
16. Oracle enterprise manager management & configuration

3.4.4. Virtualization operations, management & provisioning

1. Provision, delete, modify virtual machines
2. Manage permissions on virtual machines and VCenter
3. Resources monitoring and modifications on virtual machines
4. Track capacity growth on the VMware clusters and datastores
5. Manage virtual switches (Cisco Nexus 1000V): port profiles creation, ports monitoring, UCS manager VLAN configuration
6. Reclaim wasted resources
7. VMware Data stores management and operations
8. VMware cluster configuration, maintenance and tracking
9. High availability simulation, DRS configuration and resources reservation

3.4.5. Backup & Archiving Operations & management

1. Configure virtual machines and physical hosts backup: client installation, RMAN scripts, Exchange backup, File Server backup.
2. Monitor backup activities jobs

3. Perform restoration when required
4. Propose enhancements to backup and restoration windows
5. Perform bare metal backups and recovery
6. Configure Source One archiving rules and policies and manage Source One software
7. Configure CTA archiving rules
8. Edit file and block level storage allocation to cover them under archiving process and policy
9. Monitoring ILM cycle from Tier1 storage to Centera archiving appliance
10. Manage, configure and operate EMC Centera, that covers configuring retention and taping out

3.5. Requirements for IT & VAS Service Assurance Management

ABCD OpCo applications portfolio for IT & VAS Service Assurance are:

1. BMC Remedy
2. HP BSM
3. HP OMI
4. HP NNMI
5. HP RUM
6. HP BPM
7. HP Diagnostic
8. HP uCMDB
9. HP Site Scope
10. HP UD
11. HP operations analytics

Contractor shall provide application management services to maintain, operate, and improve ABCD OpCo specific HP and BMC Software solutions:

- Reactive Services :
  - Incident and problem management for ABCD OpCo implementation
  - Work with HP Software and BMC Product Support, ABCD OpCo, and third-party vendors to address implementation issues.
- Advisory Services :
  - Technical account management
  - Quarterly patch management
  - Periodic solution assessments, strategic planning, adoption monitoring.
- Operational & Enhancement Services :
  - Out-task maintenance and administration activities
  - A complete responsibility for ongoing solution operations
  - Implement solution enhancements.
- Below examples but not limited activities :

Solution Managed Services	Reactive Services	• Response to incidents within a defined service level objective • Solution support and defect correction
	Advisory Services	• Proactive evaluation of existing solution and recommendations for patches • Improvement of long-term stability and value of the entire solution by focusing on technical health, user adoption, and planning for continuous solution improvement

	Operational Services	• System administration, user account management • Technical support of the applications, including periodic checks to monitor application availability and response to alerts • Facilitation of the movement of changes (including defect fixes and patches) from development to test to production • Proactive review of log files and system
	Enhancement Services	• Design, development and delivery of functional enhancements to the existing solution.
New Projects	Implementation of new and major solution functionality	• Implementation of upgrades (major and minor point releases) to ensure that applications stay current and eligible for vendor technical support • Design, development, and deployment of new integrations between applications or major enhancement releases

4. Requirements for IT & VAS Applications Development

4.1. Introduction

ABCD OpCo need to focus on consumer activities accordingly ABCD OpCo need to rely on additional support to address the organization demands to adapt to the new business or technology requirements. ABCD OpCo business environment is marked by financial and growth pressure and convergence of technologies. It is a must to be able to provide a wider portfolio of consumer services, and act quickly in order to generate revenue streams.

4.2. IT & VAS Applications Development Scope

The contractor shall commit and show evidence to the following but not limited:

1. Speed & flexibility: Increased speed, productivity and flexibility of application development delivery.
2. Quality & innovation: Access to top end IT & VAS talents and better leveraged Service Provider's innovation capabilities.
3. Cost: Generating TCO improvement through Standardization, processes and tools, targeted outsourcing & off shoring, multi sourcing.
4. Efficiency: Aligned IT & VAS footprint and mode of operation with ABCD OpCo's business priorities. Improved compliance and business continuity.
5. The AD (Application Development) team shall be responsible to add functionality to the existing Applications landscape, provided ABCD OpCo has access to source code of said Applications. Application Development projects shall be categorized as Business Configuration, Simple, Medium, High and Complex based on Level of Effort and Duration.
6. AD team Services scope shall include all activities starting from High level Design until Deployment of final deliverables on the production environment.

7. Total yearly man day application development for IT & VAS shall be 6000 man day, every quarter a review will be done to share the priorities to adapt the needed resources to the business needs.. The rollover of man days is mandatory for any unused man days, man days will expire if unused, after 1 year.
8. The contractor shall fix any bug related to their development and this shall not be taken from monthly man days for development, this applies for 12 months from putting the development on production.
9. For projects that have not yet started the Development stage, ABCD OpCo has the option to utilize some of the allocated AD capacity (6,000 man-days per year).

10. The following are the requirements of the AD model but not limited:

ADM Pre-Development stage	• Demand Acceptance • Requirements Management • Quality assurance
Feasibility Study	• Solution architecture and Design • Requirements Dependencies and Impact Analysis • High-level solution design • Project Planning and Effort Estimation
Development stage	• Low level solution design • Test case definition • Business configuration & development & coding • Unit test execution report & results • Implement Software Development Lifecycle
CR Testing	• Component Unit Testing Release & Configuration Management
Post Development stage	• System integration/ Functional testing • Deployment to UAT environment • Deployment to the Live Environment
Delivery support	• Project management • Quality assurance & SLA • Configuration management • Release management • Vendor management
Continuous Improvement	• Process shall be compliant with both ITIL v3, & eTOM

4.3. Change request categories

Change Request is the medium of transferring the needed adjustments to the environment. Change Requests are mainly derived from ABCD OpCo Demand Management team, to meet any new business requirements or enhance existing operational efficiency. Change Requests (CR) contain the business requirements which need to be realized by the AD team to enable the business to offer the services to ABCD OpCo and/or internal business users. Change Requests carry a unique reference number which is used during the entire lifecycle of the CR.
CR covered under AD scope shall be categories as follows:

- Development/Customization CR
- Business Configuration CR
- Bug Fix CR
- Vendor Defects fixes testing

4.3.1. Development/Customization CR

In this category all new additional features enhancement shall be considered. New feature enhancements shall be either through additional custom development or enablement of functionalities through existing modules or code base those should go through SIT and UAT process along with contractor Managed services Operation Team and ABCD OpCo IT & Business team.

4.3.2. Business Configuration CR

In this category all business configurations changes, which require SIT and UAT shall be included under this category.

4.3.3. Bug fix CR

In this category all code fixes related to AD developments shall be considered as CR in case of expiry of warranty period (12 months) of the AD development. The bug fixes during warranty period of any custom developments shall be the responsibility of contractor AD team. In case of warranty period expired all bug fixes shall be considered as CR and covered under AD scope.
In case AD Bug fix CR identifies that it's a product related issues (Core SW or HW), Contractor Application Managed Services team shall raise the ticket to 3rd party product vendors directly.

4.3.4. Vendor Defect Fixes testing

AD test team shall perform SIT/UAT testing prior to production roll-out for SW patches which are delivered by the Software vendors and require SIT/UAT testing prior to production roll-out.

4.4. CR Complexity

It is the responsibility of Contractor to provide the Complexity Tag of any submitted CR based on the associated effort estimation and design artefacts of the CR but shall be confirmed and approved by ABCD OpCo. This complexity tag shall be defined in the Solution Design phase, and submitted to ABCD OpCo. ABCD OpCo will use the below table to calculate the Complexity Tag of any Submitted CR.

CR Category	Effort in Man Days	Time to market
Business Configuration change, BAU	1 to 15	1 week
Simple / Low CR	16 to 30	1 -2 weeks
Medium CR	31 to 60	2-4 weeks
High CR	60 to 100	4-8 Weeks
Complex CR	100 to 300	8-12 Weeks

4.5. IT & VAS Application Development monthly man day distribution

1. The table below illustrates the breakdown of the man days and effort estimated for the application development needed for the IT & VAS systems. This will be reviewed every 30-60-90 days to set the priority and needed capacity. Any additional man days needed shall be provided with a discounted rate on top of the finally agreed man day rate.
2.

IT & VAS Systems	Count of Man-day / Month
RAS, Loyalty (Seibel suite)	90
BSCS - IN	100
CRM – EBS	22
SOA – BPEL	22
Web-Logic – Online Portal	22
Stream Serve	10
Unica	22
InstaLink - Provisioning	22

Remedy BMC	10
Pay Mobile - Voucher server & etopup	10
Avaya Contact Center IVR	22
Workforce Management	10
MSDP - E///	10
SBP - Opencode	28
VAS for roaming	5
MBC/ESM - E///	66
SimTool Kit	3
OTA, ADC	3
Wataniya App	15
RA/FMS	5
Teradata Data Warehouse + ETL Informatica	5
Total	502

3. The contactor shall provide a detailed references and CVs for the following key roles:
   - Enterprise Architect
   - Business Analyst (CRM & Billing)
   - Application Development Director

5. Responsibility Matrix

The purpose of this section is to define the requirements for the roles and responsibility of the contractor.

The successful contractor must provide monitoring and management services that are aligned with ITIL v3 processes. Successful contractor must propose a clear scope of work, project plan and detailed commercial breakdown (software, Hardware, Maintenance and services) to achieve the target requirements within the six months of transition plan. More details for ABCD OpCo actual IT & VAS S tools are provided in the section “IT & VAS Service Assurance”.

5.1. Responsibility Matrix Definition

This section specifies the responsibilities applicable to all activities and deliverables and assigns responsibilities to both ABCD OpCo and successful contractor for the same.

The purpose of this responsibility matrix is to clarify the respective roles and responsibilities of both parties.

The responsibilities are defined in the table where the following interpretation of the characters shall be used:

- R = Responsible
- A = Accountable
- C = Consulted
- I = Informed

The items should be adjusted according to the result from Post Due Diligence before signing the contract and with mutual agreement of ABCD OpCo and successful contractor.

5.2. General and Service Foundation

5.2.1. General

S/N	Responsibility Description	R	A	C	I
1	IT & VAS strategy (e.g. BSS/OSS, Infrastructure strategy…)	ABCD OpCo	ABCD OpCo		
2	Telecom market strategy (e.g. branding, company vision etc.)	ABCD OpCo	ABCD OpCo		
3	Interface with ABCD OpCo customer subscriber (e.g. customer call center, billing, end user service activation etc.)	ABCD OpCo	ABCD OpCo		
4	Interface with governmental agencies such as regulatory	ABCD OpCo	ABCD OpCo		

5.2.2. Project Management Office

S/N	Responsibility Description	R	A	C	I
1	Assign program/project manager to approved program/project	Contractor	Contractor	ABCD OpCo	
2	Establish the ITO Operation governance and reporting model	Contractor, ABCD OpCo	ABCD OpCo		
3	Develop project plan for approved project	Contractor	Contractor	ABCD OpCo	
4	Establish Contractor Managed Services organization	Contractor	Contractor	ABCD OpCo	
5	Setup work interfaces between Contractor Managed Services Team and ABCD OpCo's relevant dept.	Contractor, ABCD OpCo	ABCD OpCo		
6	Setup work interfaces between Contractor Managed Services Team and other relevant Service Providers (e.g. 3rd party	Contractor	ABCD OpCo		

5.2.3. IT & VAS Tools

S/N	Responsibility Description	R	A	C	I
1	Provide ITSM Tool	ABCD OpCo	ABCD OpCo		Contractor
2	Provide application and hardware software for interfacing to the ITSM Tool	ABCD OpCo	ABCD OpCo		Contractor
3	Provide Network / Server management and monitoring systems	ABCD OpCo	ABCD OpCo		Contractor
4	Provide software distribution tool and remote support tool	ABCD OpCo	ABCD OpCo		Contractor
5	Provide IT & VAS testing tools	ABCD OpCo	ABCD OpCo		Contractor

5.2.4. Premises and Facilities

S/N	Responsibility Description	R	A	C	I
1	Provide working environment for Contractor MS team	ABCD OpCo	ABCD OpCo		
2	Provide office automation, fixed phone, internet etc for MS Team	ABCD OpCo	ABCD OpCo		
3	Provide storage and warehouse for spare parts/equipment	ABCD OpCo	ABCD OpCo		
4	Provide full and safe access to all in scope sites 7*24	ABCD OpCo	ABCD OpCo		Contractor
5	Provide data center building maintenance, equipment hosting space, site access security, electricity power, cooling system, environmental monitoring system	ABCD OpCo	ABCD OpCo		Contractor

5.2.5. IT & VAS Service Management and Operation

5.2.5.1. IT & VAS Planning

S/N	Responsibility Description	R	A	C	I
1	Provide recommendations on RFI/RFP of IT & VAS Projects	ABCD OpCo	ABCD OpCo	Contractor	
2	Select Solutions/Vendors of IT & VAS Projects	ABCD OpCo	ABCD OpCo	Contractor	
3	Negotiate Contracts with selected vendors on IT & VAS Projects	ABCD OpCo	ABCD OpCo	Contractor	

5.2.5.2. Audit

S/N	Responsibility Description	R	A	C	I
1	Support and comply with internal and external audit activities and requests on IT part	Contractor	Contractor		

5.2.5.3. Service Help Desk

S/N	Responsibility Description	R	A	C	I
1	Develop and Deploy Service Desk Process and Policy	Contractor	Contractor		ABCD OpCo
2	Provide a Service Desk service to ABCD OpCo Users.	Contractor	Contractor		ABCD OpCo
3	Process (receive, record, categorize, assign priority level, dispatch, track & monitor and closure) the calls on help	Contractor	Contractor		ABCD OpCo
4	Act as a single point of contact to communicate to the stakeholders on incident status.	Contractor	Contractor		ABCD OpCo
5	Collect Feedback and ABCD OpCo Satisfaction Survey for Contractor Managed Service.	Contractor	Contractor		ABCD OpCo
6	Handle ABCD OpCo IT service end users complaints raised from ABCD OpCo	Contractor	Contractor		ABCD OpCo

5.2.5.4. Incident Management

S/N	Responsibility Description	R	A	C	I
1	Develop and Deploy Incident Management Process	Contractor	Contractor	ABCD OpCo	ABCD OpCo
2	Approve incident management process	ABCD OpCo	ABCD OpCo		
3	Provide incident support for all levels (L1,L2,L3)	Contractor	Contractor		ABCD OpCo
4	Investigate and diagnose incidents	Contractor	Contractor		ABCD OpCo
5	Provide solution or work around for incident	Contractor	Contractor	ABCD OpCo	ABCD OpCo
6	Implement solution or work around	Contractor	Contractor	ABCD OpCo	ABCD OpCo
7	Escalate the incidents that are not resolved within SLA's	Contractor	Contractor	ABCD OpCo	
8	Acceptance solution or work around	ABCD OpCo	ABCD OpCo		

S/N	Responsibility Description	R	A	C	I
9	Review Incident Report	Contractor	Contractor		ABCD OpCo

5.2.5.5. Problem Management

S/N	Responsibility Description	R	A	C	I
1	Develop and Deploy Problem Management Process	Contractor	Contractor		ABCD OpCo
2	Approve problem management process	ABCD OpCo	ABCD OpCo		
3	Conduct major problem reviews	Contractor	Contractor	ABCD OpCo	ABCD OpCo
4	Correcting problems (error control)	Contractor	Contractor	ABCD OpCo	ABCD OpCo
5	Prepare Major Incident Report	Contractor	Contractor		ABCD OpCo
6	Resolution acceptance	ABCD OpCo	ABCD OpCo	ABCD OpCo	

5.2.5.6. Release Management

S/N	Responsibility Description	R	A	C	I
1	Develop and Deploy Release Management Processes	Contractor	ABCD M Contractor		ABCD OpCo
2	Approve release management processes	ABCD OpCo	ABCD OpCo		
3	Maintain the integrity of live environments	Contractor	Contractor	ABCD OpCo	ABCD OpCo
4	Build Releases Plan	Contractor	Contractor	ABCD OpCo	ABCD OpCo
5	Approve Release Plan	ABCD OpCo	ABCD OpCo		Contractor
6	Define and review test strategies (with entry and exit criteria) and an operational test plan methodologies	Contractor	Contractor	ABCD OpCo	ABCD OpCo
7	Build and maintain a business and technical requirements repository and test cases for systems perform system integration tests on test environment	Contractor	Contractor	ABCD OpCo	ABCD OpCo
8	Build test cases for system and integration tests	Contractor	Contractor	ABCD OpCo	ABCD OpCo
9	Deploy release to test environment	Contractor	Contractor	ABCD OpCo	ABCD OpCo

S/N	Responsibility Description	R	A	C	I
10	Provide recommendation to production promotion	Contractor	Contractor	ABCD OpCo	ABCD OpCo
11	Approve release	ABCD OpCo	ABCD OpCo		Contractor
11	Perform post-implementation review	Contractor	Contractor	ABCD OpCo	ABCD OpCo

5.2.5.7. Change Management

S/N	Responsibility Description	R	A	C	I
1	Develop and Deploy Change Management Processes	Contractor	Contractor		ABCD OpCo
2	Approve change management procedures	ABCD OpCo	ABCD OpCo		
3	Overall governance reporting and operational coordination	Contractor	Contractor		ABCD OpCo
4	Assess impact on the Change Request	Contractor	Contractor	ABCD OpCo	ABCD OpCo
5	Manage and disseminate relevant information regarding changes	Contractor	Contractor	ABCD OpCo	
6	Tracking and reporting the status of the change	Contractor	Contractor		ABCD OpCo
7	Maintain documentation on the change requests, suggested change plans and the actual changes	Contractor	Contractor		ABCD OpCo
8	Approve Change Request	ABCD OpCo	ABCD OpCo	Contractor	

5.2.5.8. Configuration Management

S/N	Responsibility Description	R	A	C	I
1	Develop and Deploy Configuration Management Processes	Contractor	Contractor ABCD OpCo	ABCD OpCo	ABCD OpCo
2	Approve configuration management procedures and policy	ABCD OpCo	ABCD OpCo		
3	Assign service identifiers for services	Contractor	Contractor	ABCD OpCo	ABCD OpCo
4	Ensure service provisioning activities are assigned, managed and tracked efficiently	Contractor	Contractor		ABCD OpCo

S/N	Responsibility Description	R	A	C	I
5	Monitor the status of service orders, provide notifications of any changes and provide management reports	Contractor	Contractor		ABCD OpCo
6	Close a service order when the service provisioning activities have been completed	Contractor	Contractor		ABCD OpCo
7	Recover specific services	Contractor	Contractor	ABCD OpCo	ABCD OpCo
8	Create and Update configuration records	Contractor	Contractor		ABCD OpCo
9	Perform audit on CMDB	Contractor	Contractor	ABCD OpCo	ABCD OpCo

5.2.5.9. Capacity Management

S/N	Responsibility Description	R	A	C	I
1	Develop and Deploy Capacity Management Procedures and Policy	Contractor	ABCD OpCo		ABCD OpCo
2	Approve capacity management procedures and policy	ABCD OpCo	ABCD OpCo		
3	Develop Capacity Plan	Contractor	Contractor		
4	Monitor utilization and analysis the trend on resource	Contractor	Contractor		ABCD OpCo
5	Review and resolve gaps on the utilization and performance against SLA/KPI and forecast capacity	Contractor	Contractor	ABCD OpCo	ABCD OpCo
6	Conduct system performance and capacity forecasting	Contractor	Contractor		ABCD OpCo

5.2.5.10.Service Level Management

S/N	Responsibility Description	R	A	C	I
1	Develop and Deploy Service Level Management Processes and Policy	Contractor	ABCD OpCo		ABCD OpCo
2	Approve service level management processes and policy	ABCD OpCo	ABCD OpCo		
2	Perform SLA base lining to establish the current benchmark	Contractor	Contractor	ABCD OpCo	
3	Monitoring and reviewing the realized service levels against the agreed/defined service levels	Contractor	Contractor	ABCD OpCo	
4	Initiate, develop and implement the service improvement plan	Contractor	Contractor	ABCD OpCo	
5	Refine the service levels as needed through Change Control Procedure	Contractor, ABCD OpCo	Contractor, ABCD OpCo		

S/N	Responsibility Description	R	A	C	I
6	Prepare Service Management report.	Contractor	Contractor		ABCD OpCo
7	Provide reporting requirements	Contractor	Contractor	ABCD OpCo	
8	Review SLA Reports	Contractor, ABCD OpCo	ABCD OpCo		
9	Plan and execute continuous service improvement plan	Contractor	Contractor	ABCD OpCo	

5.2.6. Application Management Service

5.2.6.1. Application Operation

S/N	Responsibility Description	R	A	C	I
1	Add/Delete/Modify User Account	Contractor	Contractor	ABCD OpCo	ABCD OpCo
2	Configure Application	Contractor	Contractor	ABCD OpCo	ABCD OpCo
3	Application Trouble Shooting	Contractor	Contractor		ABCD OpCo
4	Generate Management Reports	Contractor	Contractor		ABCD OpCo
5.	Data Backup and Restoration	Contractor	Contractor	ABCD OpCo	ABCD OpCo
6	Support UAT	Contractor	Contractor		ABCD OpCo
7	Response to technical queries raised by ABCD OpCo	Contractor	Contractor		ABCD OpCo
8	Patch / Upgrade Application as per agreement	Contractor	Contractor	ABCD OpCo	ABCD OpCo

5.2.6.2. Application Maintenance and Support

S/N	Responsibility Description	R	A	C	I
1	Regular health check	Contractor	Contractor		ABCD OpCo
2	Application monitoring	Contractor	Contractor		ABCD OpCo
3	Performance tuning	Contractor	Contractor	ABCD OpCo	ABCD OpCo
4	Log Archive & Purge	Contractor	Contractor	ABCD OpCo	ABCD OpCo
5	Application Reporting	Contractor	Contractor		ABCD OpCo

S/N	Responsibility Description	R	A	C	I
6	Documentation	Contractor	Contractor		ABCD OpCo
7	Problem Handling	Contractor	Contractor	ABCD OpCo	ABCD OpCo
8	Incident Handling	Contractor	Contractor	ABCD OpCo	ABCD OpCo
9	User/Permission Management	Contractor	Contractor	ABCD OpCo	ABCD OpCo
10	Application Patch Update	Contractor	Contractor	ABCD OpCo	ABCD OpCo
11	Knowledge Transfer	Contractor	Contractor		ABCD OpCo

5.2.6.3. Application development

Activity Phases	S. No	Elements of Estimation for each CR						
			Contractor	OK	OK	Contractor	Contractor	Contractor
			ADM	BU	IT Planning	QA	Operation	3rd Party Vendor Support
Demand Management	1	Requirement Definition/Inputs	I	R				
	2	Business Requirement Gathering/Clarification	I	C	R			
	3	Business Requirement Analysis	C		R			
	4	Scope of Work	C		R,A			
	5	Business Case	I	R	C			
	6	CRs decision and prioritization	C	R, A	C			
Feasibility study	7	Technical Impact Analysis	R		A			
	8	High Level Solution Design	R		A			
	9	Project Implementation Plan	R		A			
	10	Effort Estimation	R		A			
Development	11	Low Level Solution Design (LLSD)	R		A			
	12	Test Case Definition and Creation	R		A		C	
	13	Code / Configurations	R					
	14	Unit Test Execution Report and Results	R		A			

	15	Release notes document	R		A		C	
Testing	16	CR Deployment on Testing Environment	R		I	A		
	17	Test Case Definition and Creation	R		I	A	C	
	18	SIT	R		I	I	A	
	19	UAT	C	R	C	A	C	
	20	Defect Fixing during SIT/UAT	R		I	A		
Productio n Rollout	21	CR Deployment package	R		A		I	
	22	Perform Deployment on the Live System	C	I	I	I	R,A	
	23	Operation					R	
	24	L1 and L2 Support				I	R	I
	25	L3 Support (Kernel)	C		A			R
	26	L3 Support (Non Kernel)	R			A	C	C
ADM Delivery Support	27	Project Management	R		I			
	28	Quality Assurance, SLA Report	R		A	I	I	
	29	Configuration Management	R		I	I	I	I
	30	Release Management	R		I	I	I	I
	31	Vendor Management	R		C/A		I	

BU: Business User, **OP**: Operation, **QA**: Quality Assurance
OK: ABCD OpCo

5.2.7. IT & VAS Infrastructure and Platforms

5.2.7.1. Servers, Operating System, Storage, SAN, Archiving Data Base and Middleware

S/N	Responsibility Description	R	A	C	I
1	Health Check	Contractor	Contractor		ABCD OpCo

S/N	Responsibility Description	R	A	C	I
3	Capacity Management	Contractor	Contractor		ABCD OpCo
4	Archive & Purge	Contractor	Contractor	ABCD OpCo	ABCD OpCo
5	Backup, Recovery and Archiving	Contractor	Contractor	ABCD OpCo	ABCD OpCo
6	Performance Tuning	Contractor	Contractor	ABCD OpCo	ABCD OpCo
7	Performance Monitoring Reporting	Contractor	Contractor		ABCD OpCo
8	Documentation	Contractor	Contractor		ABCD OpCo
9	Network Element Request Implementation	Contractor	Contractor	ABCD OpCo	ABCD OpCo
10	Incident / Problem Handling	Contractor	Contractor		ABCD OpCo
11	User / Permission Management	Contractor	Contractor	ABCD OpCo	ABCD OpCo

5.2.8. 3rd Party Contract and Supplier Management

S/N	Responsibility Description	R	A	C	I
1	Award IT contracts to vendors for the in-scope services	Contractor	Contractor	ABCD OpCo	
2	Negotiate the IT contract terms and conditions for the in-scope services	Contractor	Contractor		ABCD OpCo
3	Monitor the contract performance for the in-scope service, and contract managed by Contractor on behalf of ABCD OpCo	Contractor	Contractor		ABCD OpCo
4	Terminate IT contracts for the in-scope service	Contractor	Contractor	ABCD OpCo	ABCD OpCo
5	Evaluate vendors for IT services and solutions for those contract managed by Contractor	Contractor	Contractor		ABCD OpCo
6	Select vendors for IT solutions and services	Contractor	Contractor	ABCD OpCo	ABCD OpCo
7	Monitor vendor performance for in-scope services	Contractor	Contractor		ABCD OpCo
8	Replace vendor of the in-scope services	Contractor	Contractor	ABCD OpCo	ABCD OpCo
9	Liaise with 3rd party vendors for maintenance and support	Contractor	Contractor	ABCD OpCo	ABCD OpCo

6. IT and VAS SLA & KPI Requirements

1. Introduction

The below sections details the expected ABCD OpCo KPIs and SLAs to be delivered by Managed Services contractor in the IT & VAS domains. The Contractor shall provide KPIs reporting on regular basis according the ABCD OpCo schedule.

The set of the KPIs present in the below sections is not exhaustive and ABCD OpCo has the right to add, change or remove the KPIs during the duration of the MS contract without additional charge.

The service shall start with the current KPIs and SLAs and over six months progress to the "Target KPIs and SLAs".

2. Key Definition

Terminology	Definition
KPIs and SLA (Service Level Agreement)	Key service targets and operational KPIs that measure the performance of the managed services
Business Delivery Channel	The application or systems that the IT & VAS department provides to other sectors, which at the end should serve the internal or external ABCD OpCo of the Company
SLA Period	The validity of the Service Level Agreement
SLA Review	Defines the frequency with which this agreement is agreed to be revised.
Problem	The underlying cause of an incident or incidents
Incident	An event that is not a part of the standard operation of a service which causes or may cause an interruption to, or reduction in the quality of that service.
Metric	The measurement that can measure the Service Level
Availability	Ability of the component or service to perform Contractor required functions at a stated instant or a stated period of time.
Response Time	The time measured between the trouble ticket being received, logged and accepted and the time of the first reply by the assigned Contractor Engineer.
Restoration Time	The time measured between the trouble ticket being received, logged and accepted and restoration of the service through work around.
Resolution Time	The time measured between the trouble ticket being received, logged and accepted and the time when the problem is definitively remedied.
Impact	Impact is defined as a measure of the business criticality of an Incident or Problem, and it is often equal to the extent to which an Incident leads to degradation of agreed service levels. Impacts are classified as: **Critical** **Major** **Minor**

Urgency	Urgency is defined as the necessary speed of solving an Incident of a certain impact. Urgency are classified as : **High** **Medium** **Low**
Severity	Severity is the result of the combination of impact and urgency levels
Planned Downtime	Downtime of a service worked out before hand and it is not a downtime caused by an incident.
Escalation	Keeping the concerned people informed at different stages of corrective maintenance (By e-mail, phone…etc.)

Business Day

Business Day is defined as normal ABCD OpCo working days, excluding public holidays, Saturday and Sunday. Normal Business Hours of ABCD OpCo is defined as from 08:00 hours till 16:00 hours (Sunday to Thursday).

Severity Level Definitions

Severity Level	Definition
Severity 1 (SL1)	Conditions that severely affect service, customer, revenue, capacity/traffic, IN, billing and maintenance capabilities and require immediate corrective action, regardless of time of day or day of the week as viewed by MS contractor, such as a loss of service that is comparable to the total loss of effective functional capability of an entire switching or transport system, a reduction in capacity or traffic handling capability such that expected loads cannot be handled and any loss of safety or emergency capability.
Severity 2 (SL2)	Conditions that seriously affect system operation, maintenance and administration, etc., and require immediate attention as viewed by the MS contractor. The urgency is less than in critical situations because of a lesser immediate or impending effect on system performance, network operations, revenue, systems' performance, reduction in any capacity/traffic measurement function, any loss of functional visibility and/or diagnostic capability, short outages equivalent to system or subsystem outages, with accumulated duration of greater than 2 minutes in any 24 hour period , or that continue to repeat during longer periods, repeated degradation or higher rate spans or connections, prevention of access for routine administrative activity, degradation of access for routine administrative operations, degradation of the system's ability to provide any required critical or major trouble notification, Any significant increase in product related customer trouble reports, corruption of system or billing
Severity 3 (SL3)	Conditions that do not significantly impair the function of the system. Problems that do not significantly impair the functioning of the system and do not significantly affect service to customers. These problems are not traffic affecting.

Impact and Urgency

The Following table shows the matrix combining the impact and urgency where SL1 is the highest priority and SL3 is the lowest priority.

		Impact		
		Critical	Major	Minor
Urgency	High	SL1	SL1	SL3

	Medium	SL1	SL2	SL3
	Low	SL2	SL3	SL3

3. Requirements

3.1. Service Coverage Hours (Service window)

Technical resources providing the service shall be "ONCALL" outside the Working Hours to attend to Severity 1 and Severity 2 incident in relation to the scope for each service. Team members from each technical discipline shall be assigned to be on call based on pre-set schedule. The on-call assigned team members shall be on standby in case of raised Severity 1 and Severity 2 incident. The **IT & VAS NOC** shall have the list of assigned on-call engineers and will contact the right on-call assigned engineer based on the incident description and symptoms. The assigned on-call engineer shall immediately attend the call through remote access facilities and will work on troubleshooting as per the incident management process.

The below table shows the detailed service window for each service:

Service	Coverage
End User Help Desk Support	During Working Hours - Outside working Hours will only provide services for emergency incident and problem - Retails and Shops On call 24x7
24 x 7 IT & VAS System Monitoring	(Shifts, 24x7)
Applications Management Service	(During Working Hours, On call, 24X7)
Infrastructure Management service	(During Working Hours, On call, 24X7)
Application Development Service	(During Working Hours, On call, 24X7)

Maintenance Window hours is defined as from 1 am to 5am.

3.2. Service Measurement Metrics (SLA)

The operational hours for each of the service is defined in section (Service Window) and the service levels measurement matrix mentioned in below Tables is going to be applied after the first six Months of operation (after ending of transition). During the first six months of operations the Managed Service contractor shall meet the final agreed SLA with ABCD OpCo.

Current SLA IT & VAS Mission Critical Systems

Type of Service	Lead-Times
Severity Level 1	

Availability 7*24	
Response	As per 3rd Party contracts
Restore	As per 3rd Party contracts
Permanent Solution	As per 3rd Party contracts
Severity Level 2	
Availability 7*24	
Response	As per 3rd Party contracts
Restore	As per 3rd Party contracts
Permanent Solution	As per 3rd Party contracts
Severity Level 3	
Availability 7*24	
Response	As per 3rd Party contracts
Restore	As per 3rd Party contracts
Permanent Solution	As per 3rd Party contracts

Target SLA IT & VAS Mission Critical Systems

Type of Service	Lead-Times	Weight	Measurement Criteria
Severity Level 1			
Availability 7*24			
Response	15 min		
Restore	4 Hours	A	Measured per incident, any each period of 15 minutes over the Lead-Times will result in a monthly service fee reduction as defined in Annex C2.
Permanent Solution	As per 3rd Party contracts		
Severity Level 2			
Availability 7*24			
Response	15 min		
Restore	8 Hours	B	Measured per incident, any each period of 15 minutes over the Lead-Times will result in a monthly service fee reduction as defined in Annex C2.
Permanent Solution	As per 3rd Party contracts		
Severity Level 3			
Availability 7*24			
Response	15 min		

Restore	16 Buisness Hours	B	Measured per incident, any each period of 15 minutes over the Lead-Times will result in a monthly service fee reduction as defined in Annex C2.
Permanent Solution	As per 3rd Party contracts		

Current SLA IT & VAS Non-Mission Critical Systems

Type of Service	Lead-Times
Severity Level 1	
Availability 7*24	
Response	As per 3rd Party contracts
Restore	As per 3rd Party contracts
Permanent Solution	As per 3rd Party contracts
Severity Level 2	
Availability 7*24	
Response	As per 3rd Party contracts
Restore	As per 3rd Party contracts
Permanent Solution	As per 3rd Party contracts
Severity Level 3	
Availability 5*8	
Response	As per 3rd Party contracts
Restore	As per 3rd Party contracts
Permanent Solution	As per 3rd Party contracts

Target SLA IT & VAS Non-Mission Critical Applications

Type of Service	Lead-Times	Weight	Measurement Criteria

Severity Level 1			
Availability 7*24			
Response	30 min	B	
Restore	8 Hours	B	Measured per incident, any each period of 15 minutes over the Lead-Times will result in a monthly service fee reduction as defined in Annex C2.
Permanent Solution	As per 3rd Party contracts		
Severity Level 2			
Availability 7*24			
Response	30 min	B	
Restore	16 Buisness Hours	B	Measured per incident, any each period of 15 minutes over the Lead-Times will result in a monthly service fee reduction as defined in Annex C2.
Permanent Solution	As per 3rd Party contracts		
Severity Level 3			
Availability 5*8			
Response	30 min	C	
Restore	As per 3rd Party contracts	C	Measured per incident, any each period of 15 minutes over the Lead-Times will result in a monthly service fee reduction as defined in Annex C2
Permanent Solution	As per 3rd Party contracts		

3.3. IT & VAS KPI

The following are the target key operational KPIs for the ABCD OpCo **IT & VAS** infrastructure and applications.

This covers 5 main categories:

1. Service Availability: Contractor shall measure service availability across all systems within the scope of the contract.
2. System response time: Contractor shall measure the response time for the top business transactions agreed during the six months of service transition.
3. Critical Revenue Impacting transactions: Contractor shall measure the systems operational performance.
4. Application development KPI
5. IT & VAS Service Management KPI

Service availability shall be measured and calculated against Incidents excluding planned outages.

Item No.	KPI Measure	KPI Target	Weight	Measurement Criteria
1	Mission Critical Systems Overall System Availability	99.999%	A	Measured monthly, for each 0.1% the Committed KPI for restore time will result in a monthly service fee reduction as defined in Annex C2.
2	Nom-Mission Critical System Overall System	99.95%	B	
			C	
3	SMS-MO Success Rate	>98%	B	
4	SMS-MT Success Rate	>98%	C	
5	SMS-MO International Success Rate	>98%	B	
6	SMS-MT International Success Rate	>98%	B	
7	SMS-MO National Success Rate	>98%	B	
8	SMS-MT National Success Rate	>98%	C	
9	RBT CPU Utilization %	< 80%	C	
10	RBT Capacity Utilization	< 80%	C	
11	RBT Call (Trunk) Success Rate	> 99%	B	
12	RBT Activation Success Rate	> 97%	B	
13	Callect call -Call (Trunk) Success Rate	> 99%	B	
14	Callect call -Call Attempts	> 200K	C	
15	Callect call -Utilization %	< 80%	C	
16	Callect call -Call Me Back Success Rate	> 99%	B	
17	RAS Transactions Success Rate	> 99.6%	B	
18	RAS Transactions MET SLA %	> 99.95%	B	
19	RAS Transactions Breached SLA %	< 0.5%	B	
20	RAS Breached SLA % completed within 10 minutes	< 0.1%	B	
21	RAS Breached SLA % completed within 60 Minutes	<0.01%	B	
22	COMTEL Total Transactions	1.000	B	
23	COMTEL Success Rate	> 95 %	B	
24	COMTEL Failed transactions	< 5 %	B	
25	MMS-MO Success Rate	>98%	B	
26	MMS-MT Success Rate	>98%	C	
27	MMS-MO International Success Rate	>98%	B	
28	MMS-MT International Success Rate	>98%	C	
29	MMS-MO National Success Rate	>98%	B	
30	MMS-MT National Success Rate	>98%	C	
31	WAP % Session Load - PUSH	> 50%	C	
32	WAP % Session Load - PULL	> 80%	C	
33	WAP All Page Browsing Success Rate	> 99%	B	
34	WAP My Homepage Browsing Success Rate	> 99%	B	
35	Anti-SPAM Capacity Utilization %	<50%	C	
36	Anti-SPAM Message Success Rate %	>98%	C	
37	IN Capacity Utilization % (CCN)	> 50%	C	
38	IN Success Rate	> 99.99%	B	
39	IN USSD Request	> 99.99%	B	

Item No.	KPI Measure	KPI Target	Weight	Measurement Criteria
40	IN Capacity Utilization % (AIR)	> 50%	C	
41	MCA Notification Success Rate	> 97%	B	
42	MSDP - Homepage Failure Rate (ABCD)	< 0.1%	C	
43	MSDP - Homepage Partial Failure Rate (ABCD)	< 1%	C	
44	MSDP -Top-40 Pages Failure Rate	< 0.2%	C	
45	MSDP - Top-40 Pages Partial Failure Rate	< 5%	C	
46	MYNET - Total Transactions	> 20K	C	
47	MYNET Success Rate	> 99%	B	

IT & VAS Service Management (ITIL V3 & eTOM Industry Standard)

Context.	KPI Measure	KPI Target	Weight	Measurement Criteria
S L A Management	*% of Met SLAs*	100%	A	For Percentage related KPI: Measured monthly, for each 0.1% not meeting the KPI target will result in a monthly service fee reduction as defined in Annex C2. For time related KPI: easured monthly, any drop for each 15 minutes over the target time will result in a monthly service fee reduction as defined in Annex C2. TBD: not yet defined, to be defined and mutually agreed at a later date.
S L A Management	*% of SLA violations*	0 %	A	
S L A Management	*% of Service Level Objectives for Met IT & VAS Process Activities*	100 %	A	
Service Help Desk	*Incident resolution time*	8 Hours	C	
Service Help Desk	*Mean time to resolve customer incident*	2 Hours	B	
Service Help Desk	*% of incidents that are reported more than once by the customer*	2 %	C	
Service Help Desk	*% of Satisfied Customers*	95 %	B	
Incident Management	*Avg Outage Duration per severity*	TBD	TBD	
Incident Management	*Number of Opened Incidents*	TBD	TBD	
Incident Management	*Number of Closed Incidents*	TBD	TBD	
Incident management	*% of problems resolved by due date*	100 %	B	
Incident Management	*Incident Resolution Time*	TBD	TBD	
Incident Management	*% of Problems By Cause Type*	95 %	C	
Incident Management	*% Maintenance Time Used For Repair*	TBD	C	
Incident Management	*Service Problem Resolution Time*	TBD	C	

Context.	KPI Measure	KPI Target	Weight	Measurement Criteria
Incident Management	*Incident Resolution Time*	TBD	C	
Incident Management	*Availability of service to customer*	100 %	B	
Change Management	*Change Success Rate*	99 %	C	
Change Management	*% of Unauthorized Implemented Changes*	0 %	C	
Incident Management	*% of Escalated Incidents*	5 %	C	
Change Management	*% of Changes Resulting in Outage*	0 %	C	
Change Management	*% of Unplanned Changes*	0 %	C	
Change Management	*Change Backlog Size*	TBD	C	
Change Management	*Number of Completed Changes*	100 %	C	
Incident Management	*Incidents Backlog Size*	TBD	TBD	
Fault Management	*Average time to create a ticket*	15 min	C	
Fault Management	*% of escalated alarms*	5 %	C	
Fault Management	*% First Level Support Resolution*	95 %	C	
Fault Management	*% Automated tickets*	90 %	C	
Fault Management	*% of Alarms reduction*	TBD	C	
Fault Management	*% of Alarms reduction based on RCA*	TBD	C	
Fault Management	*% of Alarms reduction based on SIA*	TBD	C	
Fault Management	*% tickets per element type*	100 %	C	
Fault Management	*% Automated tickets*	95 %	C	

Application Target Development SLA & KPIs

Refer to “Annex F4 -IT-VAS-Application-Development-SLA-KPI”.

7. Service Credit Scheme

The objectives of the Service Credit scheme with Contractor will be as follows:

- To drive the Contract behavior to deliver consistently good service to ABCD OpCo.
- To measure and report Service performance.
- To reflect Service performance below agreed Service Levels; and
- To be flexible enough to allow for requirements to change over time.

7.1. Requirements for Service Credit Scheme

Below ABCD OpCo requirement for Service Credit are defined in MaSSA Annex C2, Pricing Schedule:

- Existing Service Credit schemes, in current 3rd party contracts to be transferred to new Vendor, will be in place and enforced until terminated.
- ABCD OpCo shall have the right to terminate a Service Package or Service Element of the Service for continued poor performance. This requirement will be further detailed during the BAFO process.

Appendix I Glossary

No.	Abbreviation	Description
1	CM	Customer Management
2	PM	Product Management
3	EM	Enterprise Management
4	TPV	Third Party Vendor
5	CCB	Change Approval Board
6	CMDB	Configuration Database
7	CMS	Configuration Management System

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